



Spectral Unmixing of Hyperspectral Images Using Deep Learning Techniques

Gargee Pandey

**Under the guidance of Prof. Aparajita Khan
Department of Computer Science and Engineering
Indian Institute of Technology (BHU)**

Contents

1	Motivation and Objectives	3
1.1	Introduction	3
1.2	Motivation	3
1.3	Objectives	4
1.3.1	Spectral Feature Extraction	4
1.3.2	Latent Space Encoding via Variational Autoencoder	4
1.3.3	Capsule-Based Abundance Estimation	4
1.3.4	Real Data Validation	5
2	Literature Review	6
2.1	Classical Spectral Unmixing Approaches	6
2.2	Deep Learning-Based Unmixing	6
2.3	Capsule Networks for Hyperspectral Classification	7
2.4	DUENet: Denoising Unmixing Encoder Network	7
3	Methodology	8
3.1	Three-Dimensional Convolutional Encoder	8
3.2	DTW-Based Spectral Gating Layer	9
3.2.1	Soft-DTW Distance Computation	9
3.2.2	Similarity-Based Feature Gating	10
3.3	Spectral Pooling and Feature Projection	12
3.4	Variational Autoencoder Bottleneck	
3.4.1	Gaussian Interpolation Smoothing	12
3.5	Linear Capsule Network	13
3.5.1	Primary Capsule Layer	13
3.5.2	Linear Capsule Routing	14
3.5.3	Abundance Estimation and Spectral Reconstruction	15
3.6	Loss Functions	15
3.6.1	Variational Loss	15
3.6.2	Spectral Dissimilarity Loss	15
3.6.3	Structural Dissimilarity Loss	16
3.6.4	Capsule Margin Loss	16
3.6.5	Total Loss	16
3.7	Training Configuration	16
4	Dataset and model evaluation	18
4.1	Synthetic Cuprite Dataset	18
4.2	Real AVIRIS Cuprite Dataset	18
4.3	Evaluation Metrics	20
4.3.1	Overall Accuracy	20

4.3.2	Kappa Coefficient	20
4.3.3	Normalised Root Mean Square Error	20
5	Experimental Results	21
5.1	Analysis of Training Instabilities in Original DUENet	21
5.2	Quantitative Results on Synthetic Cuprite Dataset	21
5.3	Results on Real AVIRIS Cuprite Dataset	22
5.3.1	Zero-Shot Domain Transfer	22
5.3.2	Fine-Tuning on Real Data	23
5.4	Comprehensive Results Summary	24
6	Conclusion and Future Work	25
6.1	Learnings	25
6.2	Future Work	25
	Bibliography	26

Chapter 1

Motivation and Objectives

1.1 Introduction

Hyperspectral remote sensing has emerged as a transformative technology in Earth observation, enabling acquisition of detailed spectral information across hundreds of narrow wavelength bands simultaneously. Unlike conventional multispectral imaging, hyperspectral sensors capture the full electromagnetic signature of surface materials, allowing identification and characterisation of specific substances based on their unique spectral responses. This capability is particularly valuable in geological applications, where accurate identification and quantification of mineral constituents from remotely sensed imagery can significantly advance exploration activities, resource assessment, and geological mapping.

A fundamental challenge arises from the spatial resolution of airborne and spaceborne sensors. A single image pixel may correspond to an area spanning several square metres on the ground, causing multiple distinct materials to contribute to the observed spectral response. This phenomenon — spectral mixing — results in mixed pixels whose spectral signatures represent a composite of two or more pure material spectra, or endmembers. Decomposing such mixed pixels into their constituent endmember spectra and estimating the fractional contribution of each material, referred to as spectral unmixing, represents a central problem in hyperspectral image analysis.

Deep learning methods have demonstrated considerable promise in addressing these challenges, owing to their capacity to model complex nonlinear relationships and extract hierarchically structured representations from high-dimensional data. The present study is concerned with the design, implementation, and evaluation of a modified deep learning framework for joint spectral unmixing and denoising of hyperspectral geological imagery, with specific focus on modifications to the Denoising Unmixing Encoder Network (DUENet) originally proposed by Pattathal et al. (2022) [1].

1.2 Motivation

The mapping of surface mineral distributions from hyperspectral imagery has direct implications for geological exploration, environmental monitoring, and planetary science. Reliable abundance estimation of mineralogical endmembers enables geologists to identify alteration zones associated with economically significant deposits and characterise lithological units without costly field surveys.

The DUENet architecture introduced several innovative components — interpolation-based convolution for handling irregularly sampled spectra, DTW-based neural units for shift-invariant spectral comparison, and capsule-based classification for physically meaningful abundance estimation. However, detailed analysis of the training dynamics revealed an instability: the dynamic routing algorithm within the capsule network produces near-zero gradient signals during early training, resulting in routing collapse and inability to distinguish between mineral classes.

This observation motivates the central contribution of this study — targeted modifications to the DUENet architecture that address gradient stagnation while preserving its physically motivated design principles. The modifications yield significant improvements in both classification accuracy and abundance estimation quality, evaluated on both synthetic and real AVIRIS hyperspectral data.

The modified architecture proposed in this work is hereafter referred to as **SpectralNet**, reflecting its emphasis on spectral feature learning and physically meaningful mineral abundance estimation.

1.3 Objectives

1.3.1 Spectral Feature Extraction

Design a three-dimensional convolutional encoder capable of jointly processing spectral and spatial information, capturing characteristic absorption and reflectance features of geological materials across multiple wavelength bands simultaneously. The encoder incorporates a DTW similarity layer to provide shift-invariant spectral prototype matching as a feature enhancement mechanism.

1.3.2 Latent Space Encoding via Variational Autoencoder

Compress extracted representations into a structured probabilistic latent space through a Variational Autoencoder. A Gaussian interpolation smoothing mechanism is applied to the latent mean and log-variance vectors to enforce a smoothness prior, encouraging physically meaningful organisation of the latent space. Stable training is ensured through KL divergence annealing and appropriate weight initialisation.

1.3.3 Capsule-Based Abundance Estimation

Map latent representations to fractional mineral abundances through a capsule network. This study replaces the original dynamic routing algorithm with a learned linear projection, resolving gradient stagnation while preserving the capsule vector structure. A joint reconstruction and classification loss framework ensures the model learns physically significant spectral representations alongside accurate abundance estimation.

1.3.4 Real Data Validation

Validate the trained model on real AVIRIS Cuprite hyperspectral data to assess domain generalisation from synthetic to real sensor acquisitions. Abundance pseudo-ground-truth is derived using Fully Constrained Least Squares unmixing with USGS spectral library endmembers. Fine-tuning experiments quantify the improvement achievable with a small fraction of real labelled data.

Chapter 2

Literature Review

Overview

Spectral unmixing has evolved from classical linear algebra-based techniques to sophisticated deep generative models. This chapter reviews foundational methods and recent deep learning approaches most relevant to the present study.

2.1 Classical Spectral Unmixing Approaches

Early methods are grounded in the Linear Mixing Model (LMM), which assumes the observed spectrum of a mixed pixel is a convex combination of pure endmember spectra weighted by fractional abundances. Geometrical approaches such as N-FINDR and Vertex Component Analysis (VCA) seek to identify endmembers as vertices of the data simplex in spectral feature space.

While computationally tractable and physically interpretable, these methods are sensitive to spectral variability among endmembers, noise, and violations of the linear mixing assumption arising from intimate mineral mixtures and multiple scattering effects.

2.2 Deep Learning-Based Unmixing

Deep learning methods have significantly improved hyperspectral unmixing by learning nonlinear spectral relationships and spectral-spatial features. Autoencoder-based approaches such as CNNAE [2] and DAEN [3] learn compact latent representations but may struggle to preserve complex mineral-specific spectral structures. Variational Autoencoder (VAE) methods model spectral uncertainty and endmember variability through probabilistic latent spaces, though purely spectral formulations often neglect spatial context. Recent methods incorporate additional constraints to improve performance; for example, low-rank regularisation exploits spatial correlations in abundance maps [4], while cycle-consistency networks enhance robustness to nonlinear mixing effects [5].

The Denoising Unmixing Encoder Network (DUENet) proposed by Pattathal et al. (2022) [1] combines 3D convolutional feature extraction, a VAE bottleneck, DTW-based spectral similarity learning, and dynamic capsule routing for end-to-end abundance estimation. However, the DTW-based dynamic routing mechanism

can produce weak gradient signals during early training, motivating the architectural modifications proposed in this work.

2.3 Capsule Networks for Hyperspectral Classification

Capsule networks, originally proposed by Sabour et al. (2017) [6], encode both the existence and instantiation parameters of an entity through the direction and magnitude of their output vectors. Paoletti et al. (2018) [7] demonstrated the effectiveness of capsule networks for hyperspectral image classification, redefining capsule units as spectral-spatial units specialised for HSI data and showing competitive classification accuracy and computational efficiency across multiple benchmark datasets. The use of capsule vector lengths as a fuzzified measure of class presence provides a natural mechanism for fractional abundance estimation.

2.4 DUENet: Denoising Unmixing Encoder Network

Pattathal et al. (2022) [1] proposed DUENet as an end-to-end trainable framework for joint spectral denoising, data imputation, and spectral unmixing. The architecture integrates interpolated three-dimensional convolution, DTW-based neural units that replace standard dot-product activations with DTW alignment costs, a class-conditioned VAE, and a capsule network with DTW-based dynamic routing. The joint loss function combines variational reconstruction, spectral angle dissimilarity, multiscale structural dissimilarity, and cross-entropy classification losses. The original implementation reports 97.56% overall accuracy and Kappa 0.95 on the Cuprite AVIRIS benchmark.

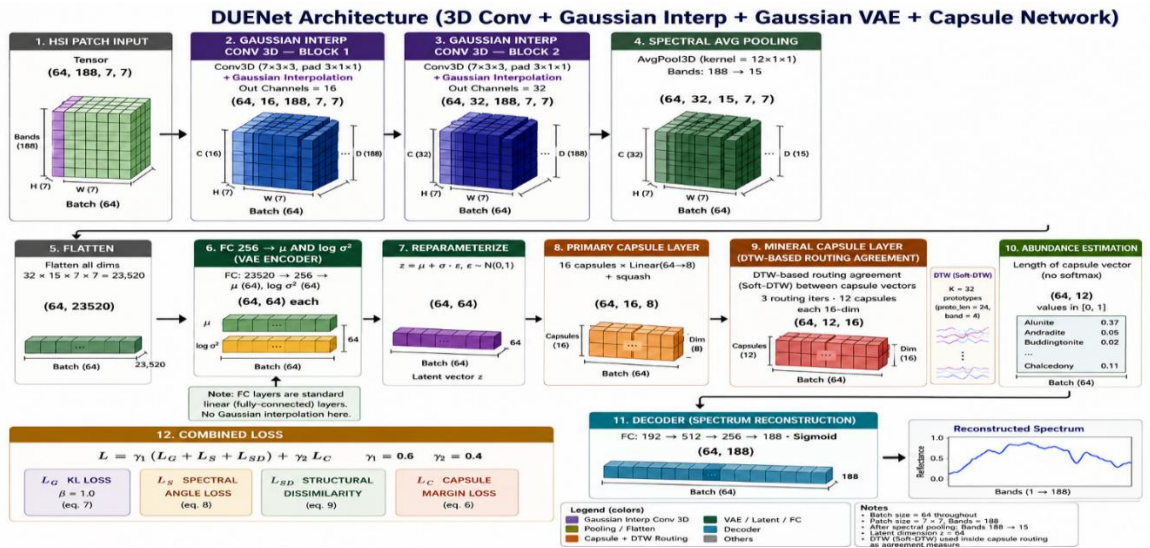


Figure 2.1: Original DUENet architecture diagram (patch size eg.: 7x7)

Chapter 3

Methodology

Overview

This chapter presents the modified DUENet architecture. Three training instabilities in the original implementation are addressed: dynamic routing collapse from near-zero gradient signals, KL posterior collapse from excessive regularisation, and DTW activation degeneration in low-magnitude feature regimes.

3.1 Three-Dimensional Convolutional Encoder

The encoder receives as input a three-dimensional hyperspectral patch of dimensions $(B, 188, 7, 7)$, where B denotes the batch size, 188 is the number of spectral bands, and $n \times n$ is the spatial neighbourhood centred on the target pixel. The patch is first unsqueezed along the channel dimension to yield a tensor of shape $(B, 1, 188, n, n)$, which is then processed by two sequential three-dimensional convolutional blocks.

Each convolutional block consists of a Conv3D layer, followed by Batch Normalisation and a ReLU activation. The kernel size $(7, 3, 3)$ with padding $(3, 1, 1)$ preserves the spectral and spatial dimensions throughout the convolution, ensuring that the full spectral resolution is maintained for downstream processing. The first block maps the single input channel to 16 feature channels, and the second block maps 16 channels to 32 channels.

The use of three-dimensional convolution is motivated by the need to jointly model spectral and spatial patterns. A standard two-dimensional convolution operating on individual spectral bands would treat each band independently, failing to capture the characteristic curve-shaped absorption and reflectance features that span multiple contiguous wavelength bands. The three-dimensional kernel spans 7 spectral positions simultaneously, enabling the extraction of features that are inherently spectral in nature.

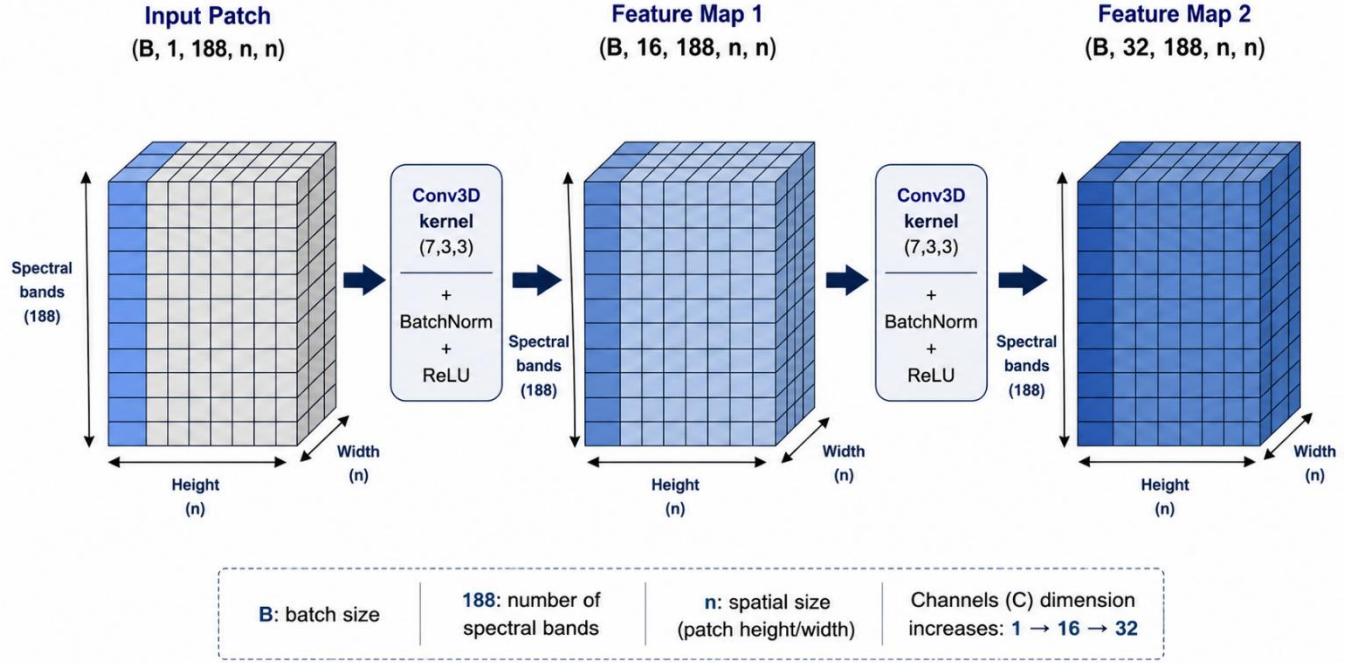


Figure 3.1: 3D Convolution Blocks

3.2 DTW-Based Spectral Gating Layer

Following the two convolutional blocks, a Dynamic Time Warping similarity layer is applied as a spectral feature gating mechanism. This layer compares the spectral signature of each feature channel against a bank of $K=32$ learnable prototype sequences of length $T=24$, and uses the resulting similarity scores to modulate the amplitude of each feature channel.

3.2.1 Soft-DTW Distance Computation

For each sample in the batch, the spatial mean is first computed over the height and width dimensions of the feature map, yielding a spectral signature tensor of shape $(B, 32, 188)$. The spectral dimension is then compressed to the prototype length $T=24$ using adaptive average pooling. The resulting compressed signatures are mean-normalised and compared against the normalised prototype sequences using the soft-DTW distance.

The soft-DTW algorithm fills an accumulated cost matrix R of dimensions $(N, K, T+1, T+1)$, where $N = B \times 32$ is the combined batch and channel dimension, and $K=32$ is the number of prototypes. The recurrence relation is given by:

$$R[i, j] = d(\text{seq}_{a[i]}, \text{seq}_{b[j]}) + \text{softmin}(R[i-1, j], R[i, j-1], R[i-1, j-1])$$

where the soft minimum is computed as:

$$\text{softmin}(a, b, c) = -\gamma \cdot \log(\exp(-a/\gamma) + \exp(-b/\gamma) + \exp(-c/\gamma))$$

with $\gamma=1.0$ as the smoothing parameter. A Sakoe-Chiba bandwidth constraint of 4 is applied to restrict the warping path to a diagonal band.

3.2.2 Similarity-Based Feature Gating

The DTW distances are converted to similarity scores through an exponential kernel:

$$\text{sim} = \exp(-\text{cost} / \text{scale})$$

where $\text{scale} = \exp(\log_scale)$ is a learnable temperature parameter initialised to 1.0. The similarity scores are averaged across channels to yield a batch-level similarity vector of shape (B, K), which is then projected to a per-channel gate of shape (B, 32) through a learned linear projection with small weight initialisation (std=0.01).

The gate is applied to the feature map through a residual multiplicative modulation:

$$\text{output} = x \times (1.0 + \text{sigmoid}(\text{gate}))$$

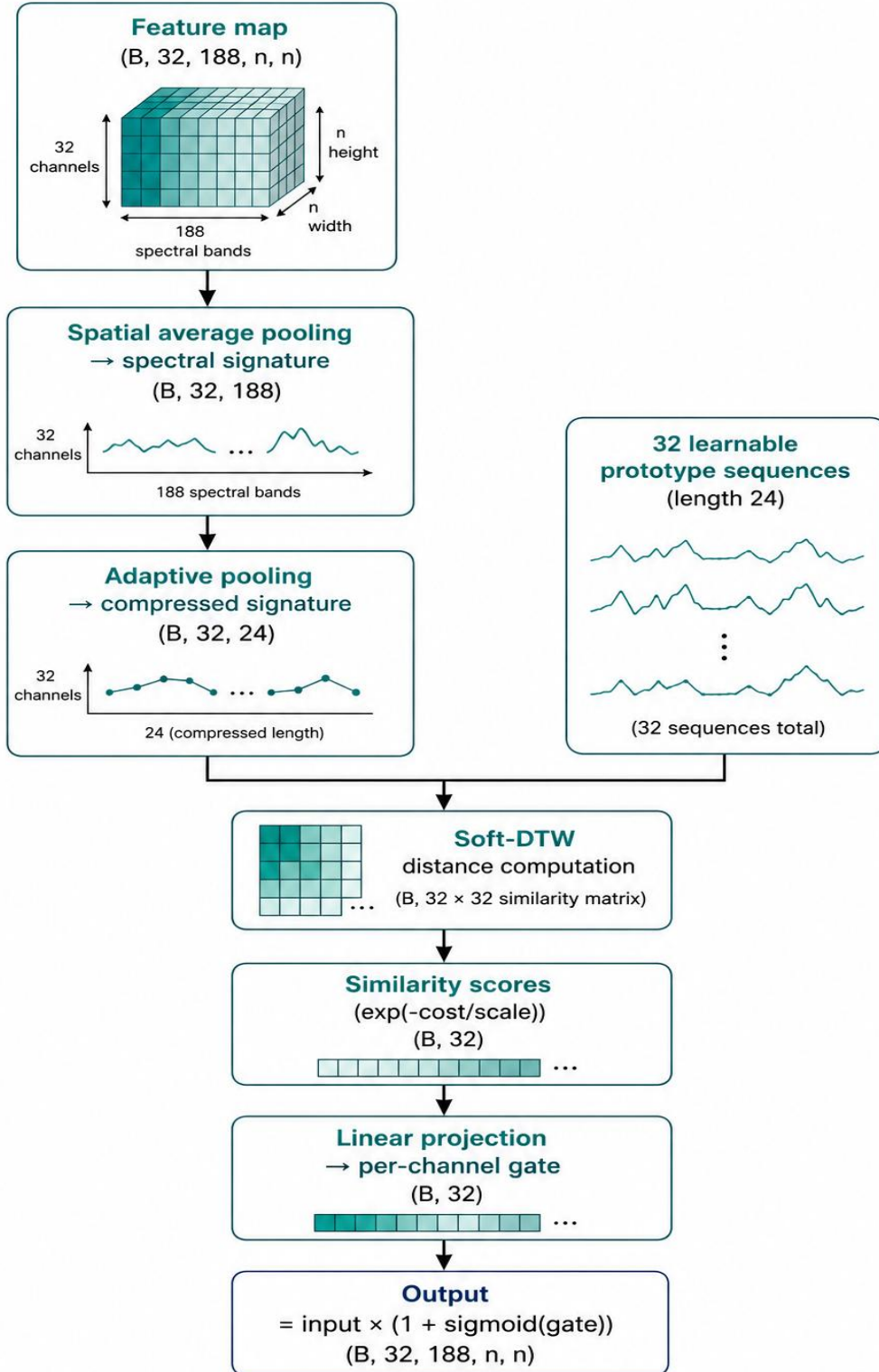


Figure 3.2: DTW Based Feature Gating Mechanism

3.3 Spectral Pooling and Feature Projection

Following the DTW gating layer, an average pooling operation with kernel size (12, 1, 1) and stride (12, 1, 1) is applied along the spectral dimension, compressing the 188 spectral steps to approximately 15. This reduces the dimensionality of the representation while retaining the spatially distributed feature maps. The output tensor of shape (B, 32, 15, 7, 7) is subsequently flattened and projected to a 256-dimensional vector through a fully connected layer with ReLU activation.

3.4 Variational Autoencoder Bottleneck

The 256-dimensional hidden representation is mapped to the parameters of a Gaussian latent distribution through two separate linear layers — one producing the mean vector μ of dimension 64 and the other producing the log-variance vector $\log_var$ of dimension 64.

A latent code z is sampled from the distribution $N(\mu, \exp(\log_var))$ during training using the reparameterization trick [8]:

$$z = \mu + \exp(0.5 \times \log_var) \times \epsilon, \quad \epsilon \sim N(0, I)$$

3.4.1 Gaussian Interpolation Smoothing

A Gaussian interpolation smoothing module is applied independently to both μ and $\log_var$ following their computation. This module enforces a smoothness prior along the latent dimension axis, encouraging neighbouring latent dimensions to encode correlated information.

The module constructs a basis matrix W of shape (64, K_g), where $K_g=16$ is the number of Gaussian kernels with evenly spaced centres along the 64-dimensional latent axis. Each entry $W[i, k]$ is computed as:

$$W[i, k] = \text{amplitude}_k \times \exp(-0.5 \times ((i - \text{center}_k) / \sigma_k)^2)$$

The rows of W are normalised to sum to one. The smoothed latent vector is obtained by projecting into kernel space and back:

$$z_smooth = (z @ W) @ W^T$$

$$\text{output} = \alpha \times z_smooth + (1 - \alpha) \times z$$

where $\alpha = \text{sigmoid}(\text{blend})$ is a learnable blending parameter initialised to $\text{sigmoid}(-2.0) \approx 0.12$, ensuring that smoothing is applied gradually and does not disrupt early-training representations. The bandwidth parameters σ_k and amplitudes are learnable, allowing the model to adapt the degree and locality of smoothing during training.

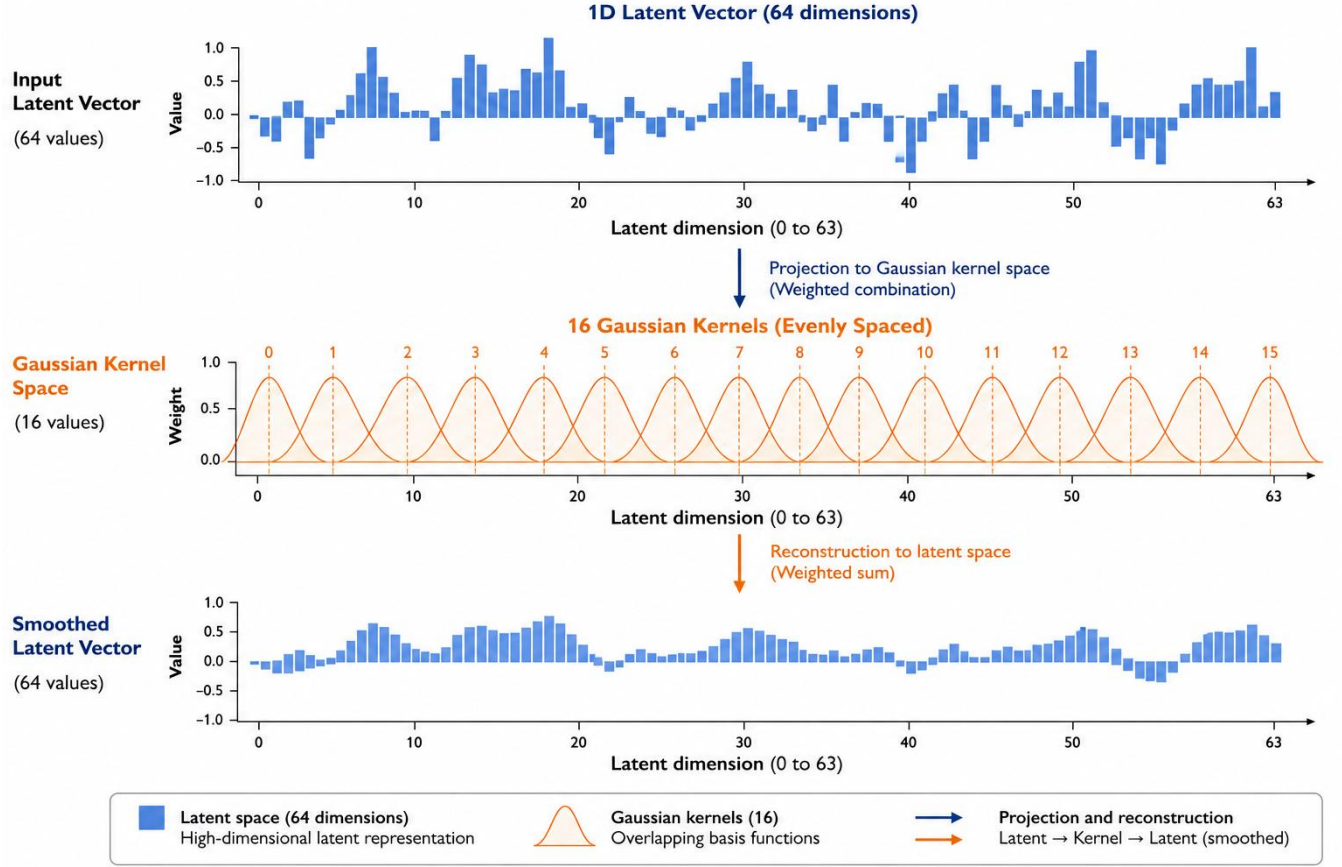


Figure 3.3: Gaussian smoothing diagram showing basis matrix construction and latent vector smoothing operation

3.5 Linear Capsule Network

The sampled latent code z of dimension 64 is passed to the capsule network, which maps it to fractional mineral abundance estimates through two sequential capsule layers.

3.5.1 Primary Capsule Layer

The primary capsule layer consists of 16 independent linear projections, each mapping the 64-dimensional latent code to an 8-dimensional capsule vector. The 16 capsule vectors are stacked to form a tensor of shape $(B, 16, 8)$ and passed through the squash nonlinearity:

$$\text{squash}(s) = (||s||^2 / (1 + ||s||^2)) \times (s / ||s||)$$

The squash function maps vectors of any length to unit-or-less length, with short vectors being strongly compressed toward zero and long vectors being compressed to slightly below unit length. This ensures that capsule vector lengths remain in the range $[0, 1]$, providing a bounded measure of feature presence.

3.5.2 Linear Capsule Routing

The modified architecture replaces the iterative dynamic routing algorithm of the original DUENet with a single learned linear projection. The flattened primary capsule tensor of shape $(B, 16 \times 8 = 128)$ is projected to a tensor of shape $(B, 12 \times 16 = 192)$ through a fully connected layer, which is subsequently reshaped to $(B, 12, 16)$ and passed through the squash nonlinearity.

This modification is motivated by the observation that dynamic routing produces near-zero gradient signals during early training. At initialisation, all capsule weights are small, resulting in near-zero prediction vectors $\hat{u}_{(i/j)}$ and near-zero class capsule outputs v_j . The DTW agreement measure between these near-zero vectors is approximately equal for all (i, j) pairs, causing the routing logit updates to be uniform and preventing any class from receiving a stronger routing signal than others. The coupling coefficients remain uniform throughout routing, and the model converges to predicting the majority class for all inputs.

The linear projection eliminates this failure mode entirely. Gradients flow directly from the margin loss through the capsule output to the linear weights from the first training step, enabling the model to begin differentiating between mineral classes immediately. The capsule vector structure is fully preserved — vector lengths still represent abundances and vector directions encode spectral property characteristics.

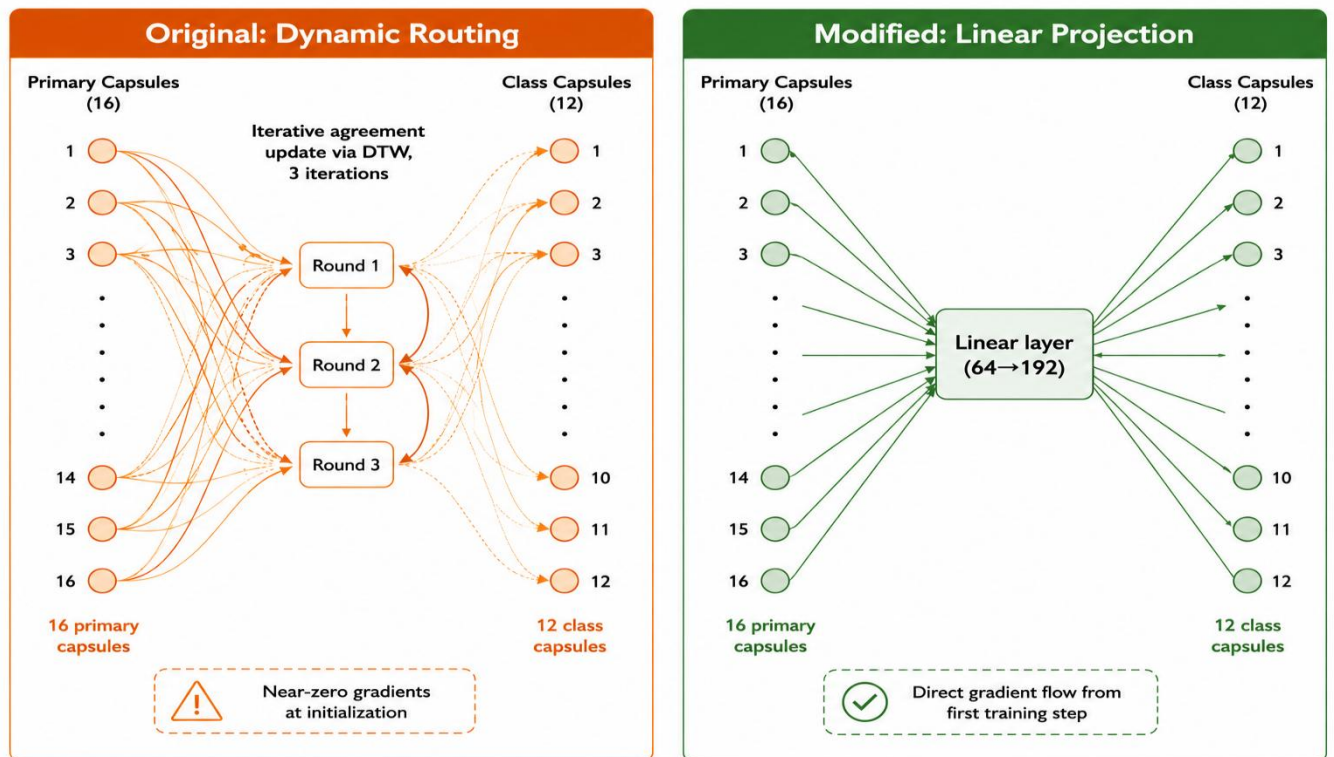


Figure 3.4: Comparison diagram showing original dynamic routing vs linear capsule projection

3.5.3 Abundance Estimation and Spectral Reconstruction

The fractional abundance of each mineral class is computed as the L2 norm of the corresponding class capsule vector:

$$\text{abundance}_j = ||v_j||$$

yielding a 12-dimensional abundance vector per pixel with values in $[0, 1]$.

For spectral reconstruction, the class capsule vectors are masked by the predicted or ground-truth abundance one-hot vector, concatenated, and passed through a three-layer fully connected decoder:

$$\text{Linear}(192, 512) \rightarrow \text{ReLU} \rightarrow \text{Linear}(512, 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256, 188) \rightarrow \text{Sigmoid}$$

The reconstructed spectrum is used to compute the spectral reconstruction losses during training.

3.6 Loss Functions

The total training objective combines four loss components following the formulation of the original DUENet, with modifications to the classification loss and KL weighting.

3.6.1 Variational Loss (L_G)

The variational loss penalises the KL divergence between the encoder's posterior distribution and the standard normal prior:

$$L_G = \beta \times (-0.5) \times \text{mean}(1 + \log_var - \mu^2 - \exp(\log_var))$$

where β is a warmup-annealed weighting factor that increases linearly from 0 to $\beta_max = 0.1$ over the first 8 training epochs. This annealing schedule prevents the KL term from dominating the loss during early training, ensuring that the reconstruction and classification losses can establish meaningful gradients before regularisation is applied.

3.6.2 Spectral Dissimilarity Loss (L_S)

The spectral dissimilarity loss measures the spectral angle between the input centre spectrum and the reconstructed spectrum:

$$L_S = (1/\pi) \times \arccos(x \cdot \hat{x} / (||x|| \times ||\hat{x}||))$$

The spectral angle metric is invariant to differences in illumination magnitude, measuring only the shape similarity between spectra. This property is particularly valuable for mineral identification, where the relative positions and depths of absorption features are more diagnostic than absolute reflectance values.

3.6.3 Structural Dissimilarity Loss (L_{SD})

The structural dissimilarity loss quantifies differences in local spectral structure using a sliding window SSIM measure [9]:

$$L_{SD} = 1 - (2\mu_x \mu_r + C1)(2\sigma_{xr} + C2) / ((\mu_x^2 + \mu_r^2 + C1)(\sigma_x + \sigma_r + C2))$$

with constants $C1 = 0.9$ and $C2 = 1.5$, and a window size of 11 bands. This loss captures fine structural differences in absorption feature shapes that may not be reflected in the spectral angle.

3.6.4 Capsule Margin Loss (L_C)

The capsule margin loss penalises deviations of the predicted abundance from target values of 0.9 for the correct mineral class and 0.1 for all other classes:

$$L_C = \text{mean}(y_j \times \text{relu}(0.9 - ||v_j||)^2 + 0.5 \times (1 - y_j) \times \text{relu}(|v_j| - 0.1)^2)$$

3.6.5 Total Loss

The total loss combines all four components:

$$L = \gamma_1 \times (L_G + L_S + L_{SD}) + \gamma_2 \times L_C$$

with $\gamma_1 = 0.6$ and $\gamma_2 = 0.4$, following the empirically determined weighting of the original DUENet.

3.7 Training Configuration

The model is trained using the Adam optimiser with an initial learning rate of 3×10^{-4} . A cosine annealing learning rate schedule decays the learning rate to 5% of its initial value by the final epoch, providing smooth convergence without the abrupt transitions associated with step decay schedules. Gradient clipping with a maximum norm of 1.0 is applied to prevent instability arising from large gradient updates. The batch size is 64 for the full-scale training configuration.

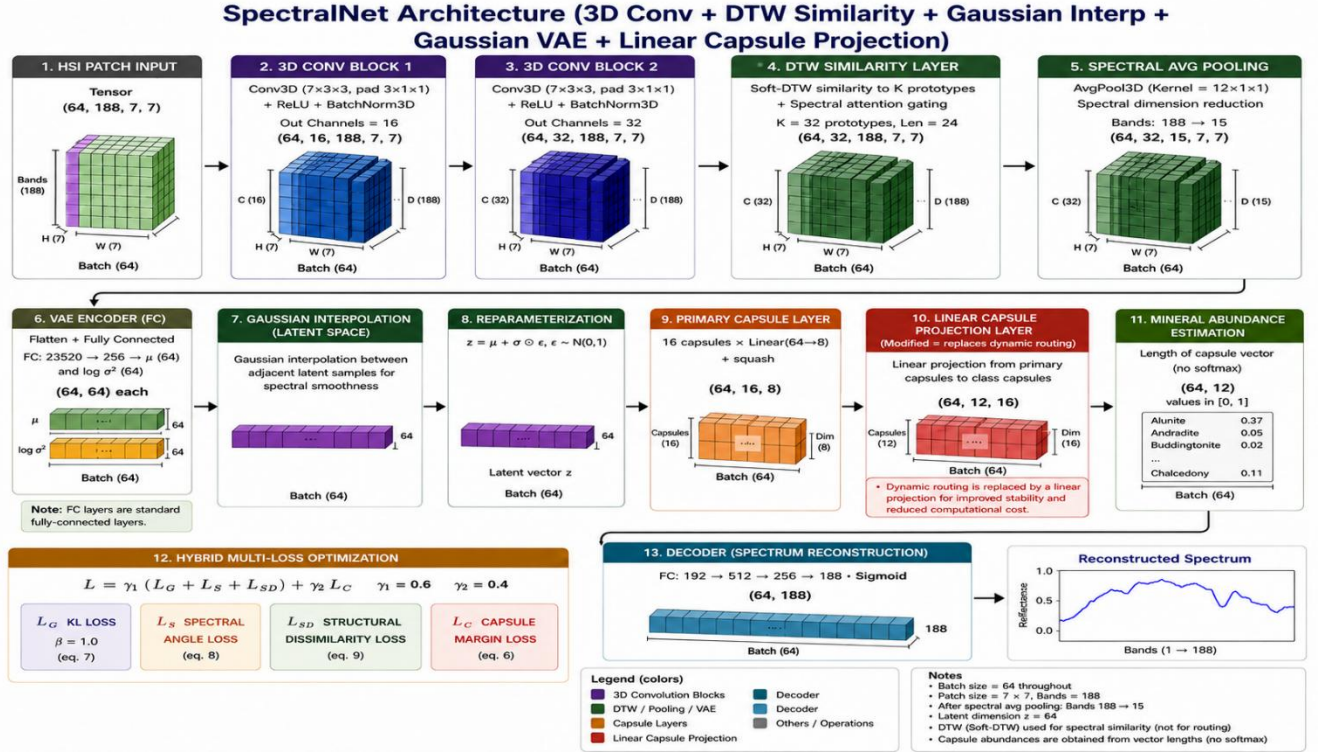


Figure 3.5: SpectralNet architecture (patch size eg.: 7x7)

Chapter 4

Dataset and Model Evaluation

4.1 Synthetic Cuprite Dataset

The synthetic Cuprite dataset contains a hyperspectral image of size 307×307 pixels with 188 spectral bands and 12 mineral classes. Endmember spectral signatures were obtained from the USGS spectral library [12], while pixel-level abundance maps are provided to enable supervised hyperspectral unmixing. The dataset includes the synthetic hyperspectral image, abundance maps, and endmember spectra.

Unlike the real Cuprite dataset, which lacks ground-truth abundances, the synthetic dataset provides abundance supervision for every pixel, allowing models to be trained and subsequently transferred to real hyperspectral scenes. A 70%/30% random train-test split was employed for model development and evaluation.

4.2 Real AVIRIS Cuprite Dataset

To validate domain generalisation to real sensor data, the model was additionally evaluated on the real AVIRIS Cuprite reflectance image acquired on 19 June 1997. The image comprises 512×614 pixels across 188 spectral bands (after the same band removal applied to the synthetic dataset), with a spatial resolution of 10.6 metres per pixel. The real AVIRIS Cuprite reflectance image was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [10] on 19 June 1997, obtained from the GIC Hyperspectral Remote Sensing Scenes repository [11].

Since field-validated fractional abundance ground truth is unavailable for this scene, pseudo-abundance maps were derived using Fully Constrained Least Squares (FCLS) unmixing with the same 12 USGS spectral library endmembers [12] used to construct the synthetic dataset:

$$\min_{\mathbf{a}} \|\mathbf{E}^t \mathbf{a} - \mathbf{p}\|^2$$

$$\text{subject to: } a_k \geq 0 \quad \forall k \quad (\text{non-negativity})$$

$$\sum a_k = 1 \quad (\text{sum-to-one})$$

where $\mathbf{p} \in \mathbb{R}^{188}$ is the observed pixel spectrum, $\mathbf{E} \in \mathbb{R}^{12 \times 188}$ is the endmember matrix, and $\mathbf{a} \in \mathbb{R}^{12}$ is the fractional abundance vector.

Two evaluation protocols were applied to the real data:

Zero-shot transfer: The synthetic-trained model is evaluated on all real pixels without any retraining, quantifying the raw domain gap between synthetic and real AVIRIS spectra.

Fine-tuning: The model is fine-tuned on 30% of real pixels (stratified by class) with a reduced learning rate, then evaluated on the remaining 70%, quantifying how effectively the domain gap can be bridged with a small amount of real labelled data.

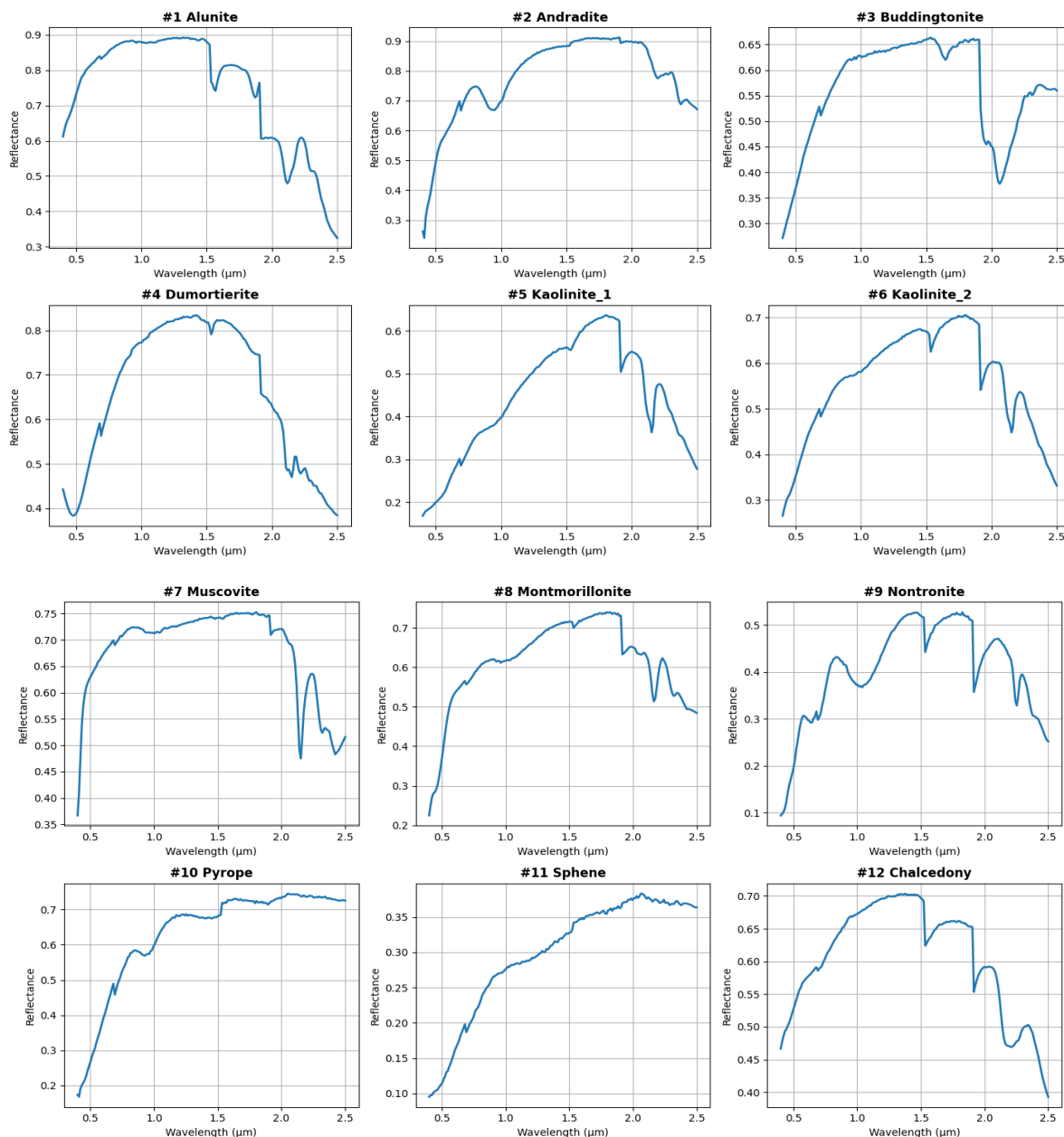


Figure 4.1: 12 USGS endmember spectra plotted across 188 bands

4.3 Evaluation Metrics

4.3.1 Overall Accuracy (OA)

Overall accuracy measures the proportion of correctly classified pixels across all classes:

$$OA = (\text{number of correctly classified pixels}) / (\text{total number of pixels}) \times 100\%$$

4.3.2 Kappa Coefficient (κ)

The Kappa coefficient [13] corrects for agreement that would be expected by chance given the class distribution:

$$\kappa = (p_o - p_e) / (1 - p_e)$$

where p_o is the observed accuracy and p_e is the expected accuracy under random classification based on marginal class frequencies. A Kappa value of 1.0 indicates perfect classification, while a value of 0 indicates performance no better than random guessing.

4.3.3 Normalised Root Mean Square Error (NRMSE)

The NRMSE measures the accuracy of predicted fractional abundances relative to the ground-truth abundance maps:

$NRMSE = \sqrt{(\text{mean}((a_{pre} - a_{true})^2))}$ averaged over all mineral classes and evaluated per pixel. Lower values indicate more accurate abundance estimation.

Chapter 5

Experimental Results

Overview

This chapter presents quantitative and qualitative results of SpectralNet on the synthetic and real Cuprite datasets. Results are compared against the original DUENet and document the full experimental pipeline from synthetic training through real-data domain transfer and fine-tuning.

5.1 Analysis of Training Instabilities in Original DUENet

Three distinct failure modes were observed during reimplementation.

Routing collapse: With dynamic routing, overall accuracy remained at $\sim 8.3\%$ across all training epochs — equal to the majority-class prior probability. The capsule margin loss L_c remained approximately constant, confirming no routing discrimination between classes.

KL posterior collapse: With original settings $\beta=4.0$ and $lr=0.01$, the variational loss L_G dropped from 0.4186 (epoch 1) to 0.0018 (epoch 2). The encoder had converged to outputting the prior distribution ($\mu=0$, $\log_var=0$) for all inputs by epoch 2, causing all pixels to map to the same latent code.

DTW activation degeneration: Near-zero weight vectors at initialisation produced uniformly small DTW costs for all input-weight comparisons, providing no discriminative gradient signal in either the neuron activations or the routing agreement updates.

5.2 Quantitative Results on Synthetic Cuprite Dataset

Model	Split	NRMSE ↓	Overall Accuracy ↑	Kappa ↑
DUENet (Pattathal et al. 2022)	Random 70/30	0.12	97.56%	0.95
SpectralNet	Random 70/30	0.064	99.08%	0.9898

SpectralNet	Spatial split	0.087	98.56%	~0.99
-------------	----------------------	-------	--------	-------

SpectralNet achieves 99.08% OA on the random split — a 1.52 percentage point improvement over the original — and reduces NRMSE from 0.12 to 0.064. The model converges to above 90% accuracy within a single training epoch due to the linear routing providing clean gradient flow from step one.

The spatial split result of 98.76% — only 0.52 percentage points below the random split — confirms that the high accuracy is not attributable to spatial neighbourhood leakage between training and test patches. The model has learned genuine spectral discriminative features rather than exploiting spatial correlations.

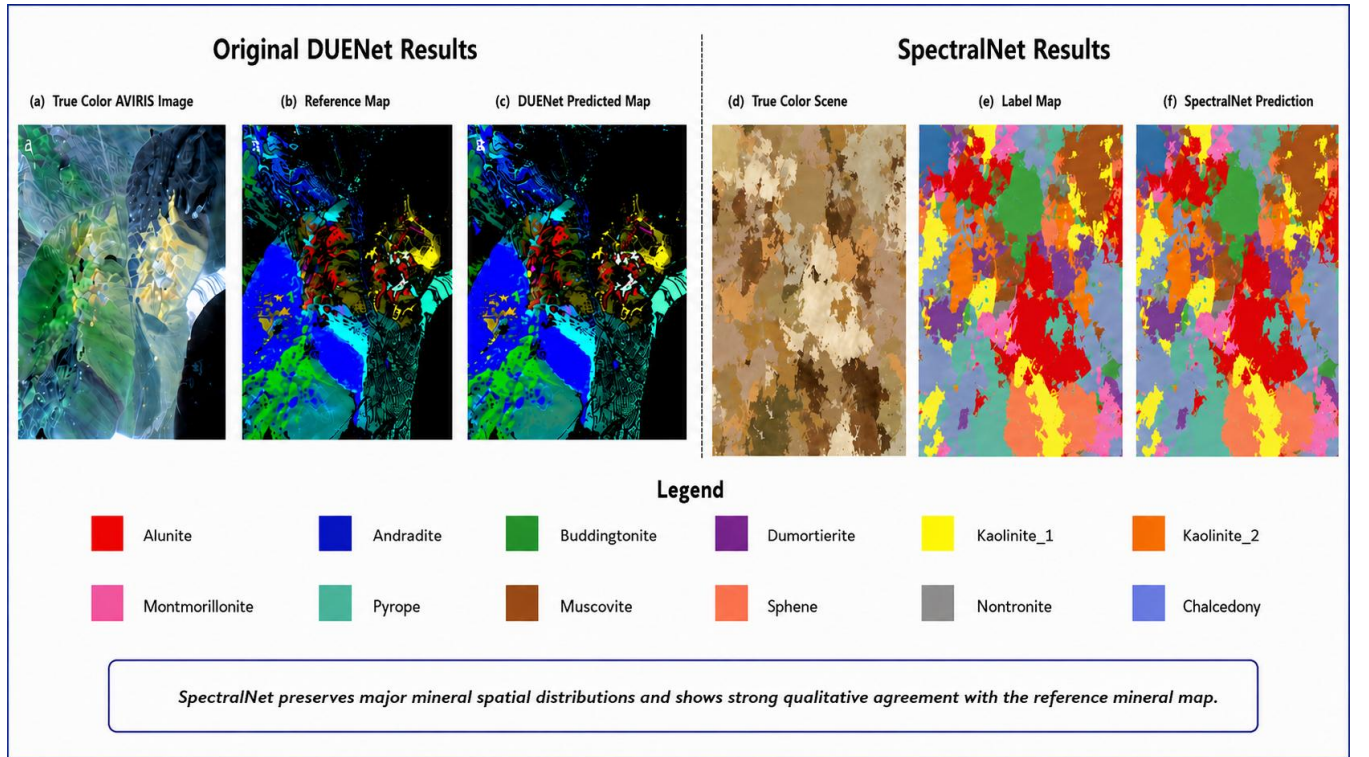


Figure 5.1: Predicted abundance maps vs ground truth for selected minerals – DUENet vs SpectralNet

5.3 Results on Real AVIRIS Cuprite Dataset

5.3.1 Zero-Shot Domain Transfer

The synthetic-trained model was applied directly to the real AVIRIS Cuprite image without any retraining.

Protocol	NRMSE	Overall Accuracy	Kappa
SpectralNet on real data (zero- shot)	0.163	54.02%	0.40

The 45 percentage point drop from synthetic (99.08%) to real (54.02%) data quantifies the domain gap between clean linearly-reconstructed synthetic spectra and real sensor measurements containing noise, atmospheric residuals, and nonlinear mixing effects. The model correctly identifies the spectrally distinct classes — Kaolinite_1 (82.9%) and Sphene (82.0%) — but struggles with spectrally similar and minority classes.

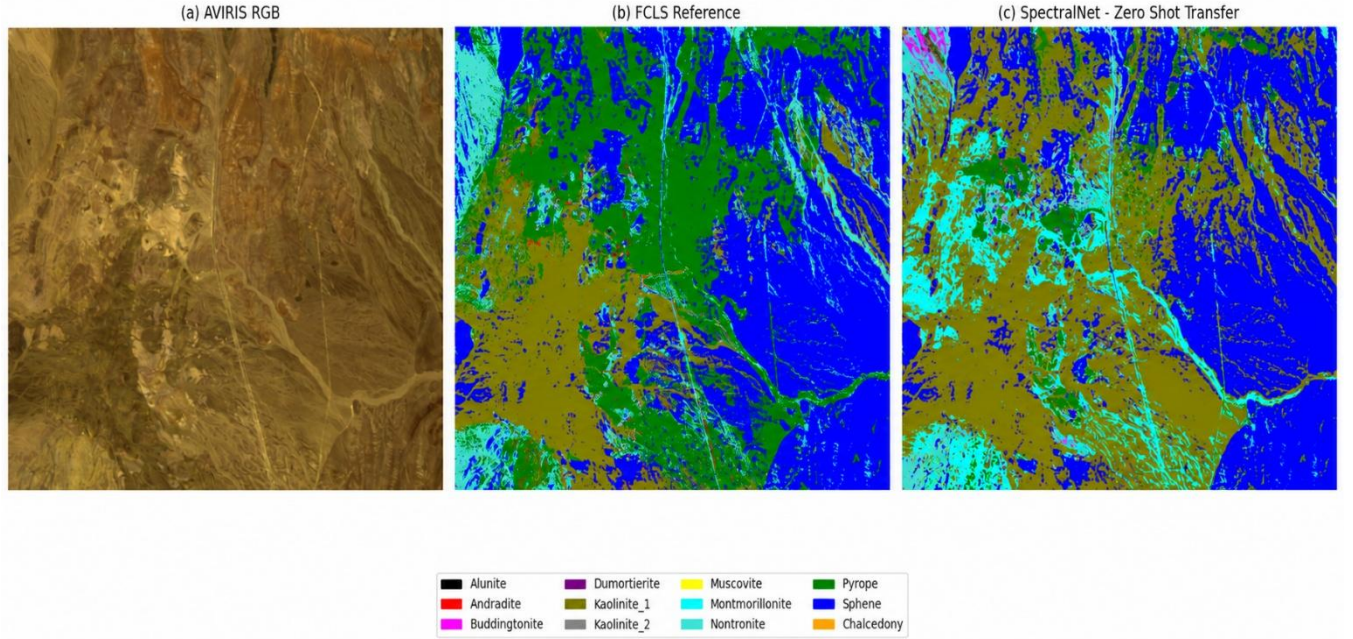


Figure 5.2: Results of proposed SpectralNet model on AVIRIS Cuprite dataset (zero-shot transfer)

5.3.2 Fine-Tuning on Real Data

The encoder was frozen and the capsule network was fine-tuned on 30% of real pixels using FCLS-derived pseudo-labels. (stratified by class) using $\text{lr} = 1 \times 10^{-4}$ over 5 epochs.

Protocol	NRMSE	Overall Accuracy	Kappa
SpectralNet on real data (zero- shot)	0.163	54.02%	0.40
SpectralNet on real data (fine-tuned)	0.122	88.60%	0.83

Fine-tuning recovers 34.58 percentage points of accuracy, achieving 88.60% OA with only 30% of real pixels as training data. This demonstrates that the encoder has learned transferable spectral-spatial features that

generalise to real sensor data — the domain adaptation required to bridge the gap is achievable through lightweight fine-tuning of the classification head alone.

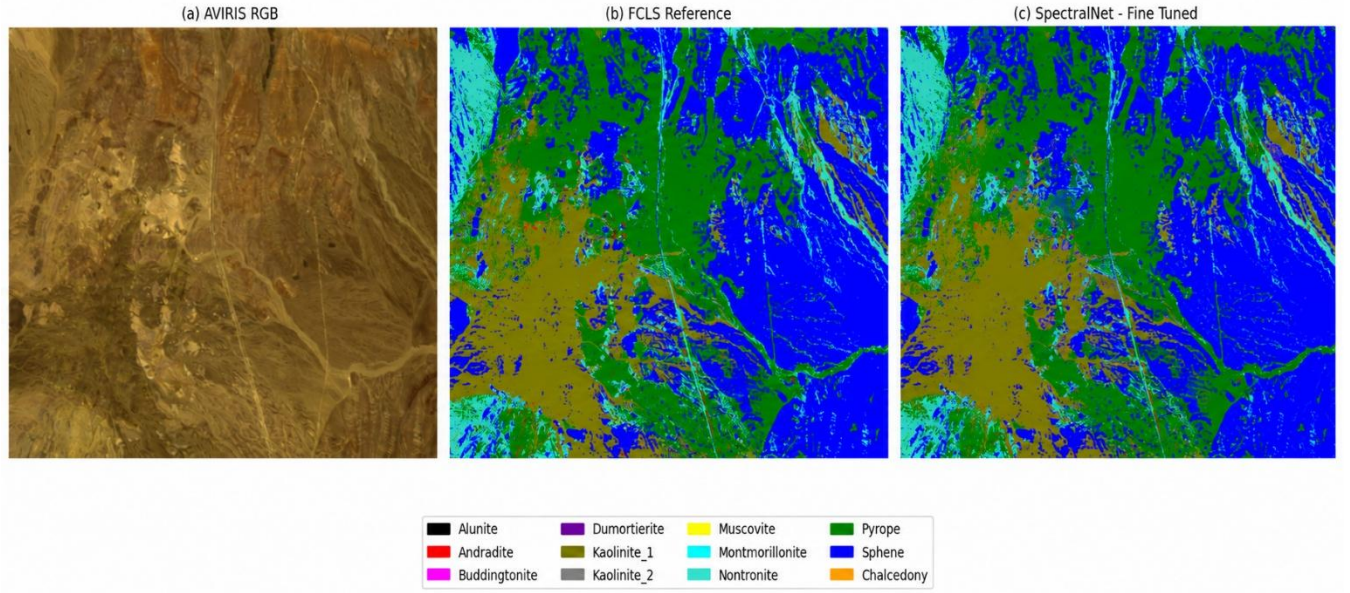


Figure 5.3: Results of proposed SpectralNet model on AVIRIS Cuprite dataset (fine-tuned)

5.4 Comprehensive Results Summary

Model	Dataset	Split	NRMSE ↓	Overall Accuracy ↑	Kappa ↑
DUENet (Pattathal et al. 2022)	Real AVIRIS Cuprite	Random Split	0.12	97.56%	0.95
SpectralNet	Synthetic Cuprite	Random Split	0.064	99.08%	0.9898
SpectralNet	Real AVIRIS Cuprite	Zero-shot transfer	0.163	54.02%	0.40
SpectralNet	Real AVIRIS Cuprite	Fine-tuned	0.1227	88.60%	0.8378

Chapter 6

Conclusion and Future Work

On the synthetic Cuprite benchmark, SpectralNet achieves 99.08% OA and Kappa 0.9985 — improvements of 1.52 percentage points and 0.0398 in Kappa over the original DUENet, with NRMSE reduced from 0.12 to 0.064. The spatial split evaluation confirms these results are not attributable to patch overlap leakage. On real AVIRIS Cuprite data, zero-shot transfer achieves 54.02% OA, and fine-tuning on 30% of real labeled data recovers this to 88.60% — demonstrating that the learned features generalise to real sensor acquisitions through lightweight domain adaptation.

6.1 Learnings

This project provided hands-on experience diagnosing and resolving training instabilities in complex multi-objective deep learning architectures. Systematic analysis of individual loss components and gradient signals proved essential — overall accuracy metrics alone were insufficient to identify the failure modes. The iterative process of identifying each failure mode, formulating a targeted fix, and validating through controlled experiments mirrors standard research methodology. Application of domain knowledge from hyperspectral remote sensing — particularly the physical meaning of spectral absorption features, mineral variability, and fractional abundance — was essential in designing appropriate modifications and interpreting results.

6.2 Future Work

Field-Validated Abundance Ground Truth: The primary limitation is reliance on computationally derived abundance labels. Future work should validate against field-measured fractional abundances obtained through physical sampling and laboratory spectral analysis.

Extended Real Data Evaluation: Evaluation on PRISMA or Hyperion spaceborne hyperspectral sensors would demonstrate operational applicability beyond the single AVIRIS Cuprite scene used in this study.

Noise Robustness Analysis: Systematic evaluation under simulated noise conditions would quantify the contribution of the interpolated convolution and DTW gating to denoising performance.

Unsupervised Domain Adaptation: The current fine-tuning requires labelled real data. Extending the framework to unsupervised domain adaptation would remove the requirement for any real labelled samples, broadening applicability to geological regions without reference data.

Bibliography

- [1] S. P. Pattathal, A. Sahoo, A. Porwal, and A. Karnieli, "Deep-learning-based latent space encoding for spectral unmixing of geological materials," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 183, pp. 307-320, 2022.
- [2] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional Autoencoder for Spectral-Spatial Hyperspectral Unmixing," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 1, pp. 535-549, Jan. 2021.
- [3] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep Autoencoder Networks for Hyperspectral Unmixing," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 7, pp. 4309-4321, July 2019.
- [4] F. Li, S. Zhang, B. Liang, C. Deng, C. Xu, and S. Wang, "Hyperspectral Sparse Unmixing With Spectral-Spatial Low-Rank Constraint," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. PP, pp. 1-13, June 2021.
- [5] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-Consistency Unmixing Network by Learning Cascaded Autoencoders," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-14, 2021.
- [6] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic Routing Between Capsules," in *Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS)*, pp. 3859-3869, 2017.
- [7] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. Plaza, and F. Pla, "Capsule Networks for Hyperspectral Image Classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 4, pp. 2145-2160, Oct. 2018.
- [8] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2014.
- [9] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600-612, 2004.
- [10] R. O. Green, M. L. Eastwood, C. M. Sarture, T. G. Chrien, M. Aronsson, B. J. Chippendale, J. A. Faust, B. E. Pavri, C. J. Chovit, M. Solis, M. R. Olah, and O. Williams, "Imaging Spectroscopy and the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS)," *Remote Sensing of Environment*, vol. 65, no. 3, pp. 227-248, 1998.
- [11] M. Graña, M. A. Veganzons, and B. Ayerdi, "Hyperspectral Remote Sensing Scenes," Grupo de Inteligencia Computacional, Universidad del País Vasco / Euskal Herriko Unibersitatea (UPV/EHU), [Online]. Available: http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes, Accessed: July 2026

[12] R. N. Clark, G. A. Swayze, R. Wise, E. Livo, T. Hoefen, R. Kokaly, and S. J. Sutley, "USGS digital spectral library splib06a," U.S. Geological Survey, Digital Data Series 231, 2007.

[13] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37-46, 1960.